

Grade 7
Unit 3
Lesson 6 Practice

Name _____
Date _____
Period _____

1. Delaney purchased a new boat for \$61,014.00. She read that the value of new boats change by $\frac{-2}{19}$ in the first year. What is the value of Delaney's boat after the first year?
2. Delaney learns that the value of her boat will continue to change by $\frac{-3}{10}$ of its value each year after the first year. Using the value after year one from Question 1, what is the value of Delaney's boat at the end of year 2?
3. An Orca whale is swimming $-176\frac{1}{4}$ ft below sea level when he notices the rest of his pod above him at the surface. Orca whales can swim at a rate of 120.52 feet per hour. How long, in hours, will it take the Orca to reach his pod?
4. A candle store sells the 3-wick candles for $2\frac{1}{5}$ times what they paid for them from the warehouse. The candles are purchased from the warehouse at \$5.65. How much profit does the candle store make off each candle?
5. After 3 months, the price of the candles is reduced by $\frac{1}{5}$ of the original store price. Using the sale price from Question 4, what is the price of the candles after the 3 month reduction?



6. Marie's office has a parking garage that charges \$10.50 each day for parking. Marie's employer provides her with a parking card that has \$200 each month. When Marie parked today, the card indicated it had \$116 left on the card. How many days has Marie used the card for parking this month?
7. Tanner is traveling on his first flight to visit family. $8\frac{4}{5}$ minutes into the flight, the pilot announces the plane is currently at an altitude of 3740 feet. At that rate, how long will it take to reach a cruising altitude of 32,400 feet?
8. A penguin sitting on an iceberg notices a fish swimming below. The fish is 10.3 ft below sea level. The penguin dives and reaches the fish in $\frac{7}{15}$ of a minute. What was the rate of descent the penguin made?
9. The temperature in North Dakota today was -8.8°C . Over the next 3 weeks, the temperature will increase an average of 3.2°C each week. What will the temperature be at the end of the 3 weeks?
10. Lucy buys her cats a bag of food that is 35.3 ounces. Each of Lucy's cats eat $\frac{9}{15}$ ounces a day. If she has 2 cats, how long will the bag of cat food last Lucy?